Lab 6: Code and Branch Coverage Analysis

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Section: D	**Assigned Module:** bookstore_trade_in.py

1. Code Coverage Analysis

Commands: coverage run -m pytest CodeCoverageTest_bookstore_trade_in.py | coverage report -m

- **Implementation:** 12 test cases in CodeCoverageTest_bookstore_trade_in.py covering all 57 statements.
- **Results:** 57/57 statements covered, 0 missed -> **100% Line Coverage** (Requirement: >= 90%). 12 passed in 0.06s.

Test Case	Target Statements / Logic	Expected Outcome
test_single_paperback_good	Paperback ($8 base, good condition)	Payout: $8.0, COMPLETED
test_invalid_batches_input	Guard clause: batches not a list (lines 2-3)	ValueError raised
test_unapproved_batch_skipped	Intake approval guard (intake_approved=False)	Batch skipped, payout $8.0
test_unsupported_format_rejected	Unknown format rejection (lines 22-23)	Batch rejected, payout $8.0
test_missing_pages_rejected	Condition missing_pages > 0 (lines 33-34)	Book rejected, payout $8.0
test_textbook_format_and_counterfeit	Textbook format ($40) & counterfeit break	Payout: $0.0, ZERO_CREDIT
test_hardcover_mint_credit_bonus	Hardcover ($15) * mint (1.25) + $10 bonus	Payout: $28.75, COMPLETED
test_paperback_poor_and_counterfeit	Poor condition (0.4x) + non-textbook counterfeit	Payout: $3.20, COMPLETED
test_audit_rejected_status	Threshold rejected_items > 4 (line 55)	Status: AUDIT_REJECTED
test_scanner_no_data	Empty scanner matrix (lines 62-63)	Returns NO_DATA
test_scanner_recalibrate	Error rates > 5 triggering recalibration	Returns RECALIBRATE
test_scanner_calibrated	Error rates <= 5 resetting streak	Returns CALIBRATED

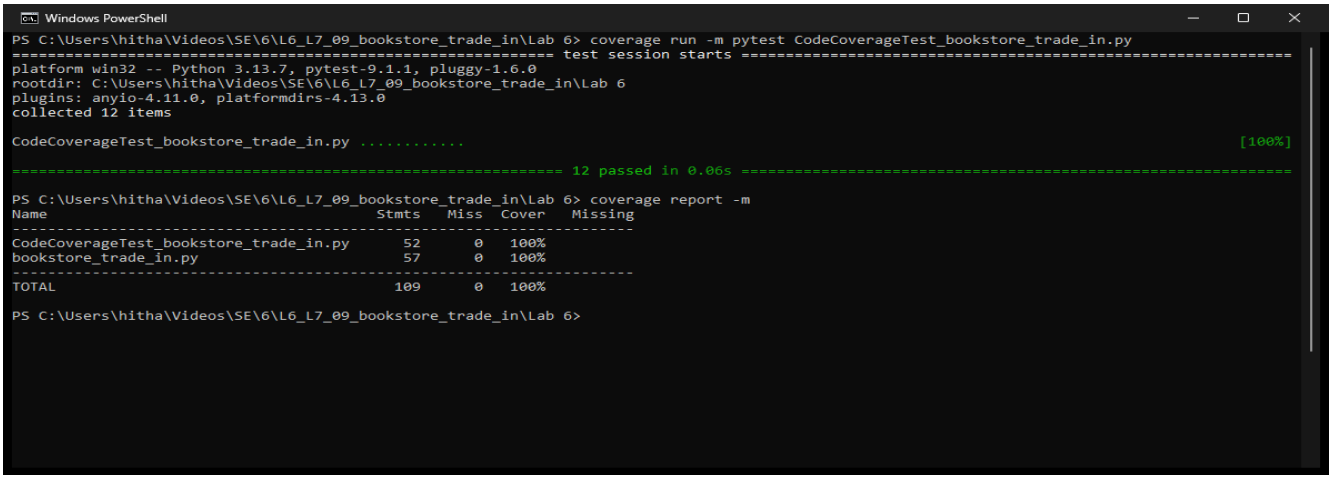


Figure 1: Code Coverage Terminal Output (100% Line Coverage)

2. Branch Coverage Analysis

Commands: coverage run --branch -m pytest BranchCoverageTest_bookstore_trade_in.py | coverage report -m

- **Implementation:** 24 test cases in BranchCoverageTest_bookstore_trade_in.py testing True and False on all decisions.
- **Results:** 40/40 branches covered, 0 missed -> **100% Branch Coverage** (Requirement: >= 90%). 24 passed in 0.07s.

Decision Point / Condition	Line	True Branch Tested	False Branch Tested
not isinstance(batches, list)	2	String input -> raises ValueError	Valid list -> execution proceeds
store_credit_bonus < 0	2	Negative bonus -> raises ValueError	Valid bonus -> execution proceeds
not batch.get('intake_approved')	9	approved=False -> batch skipped	approved=True -> batch processed
format_type == 'textbook'	15	Textbook -> base price $40.0	Other format -> checks hardcover
format_type == 'hardcover'	17	Hardcover -> base price $15.0	Other format -> checks paperback
format_type == 'paperback'	19	Paperback -> base price $8.0	Unknown format -> rejected
pages_missing > 0	32	Pages missing -> item rejected	Pages missing == 0 -> checks counterfeit
is_counterfeit and textbook	38	Counterfeit textbook -> break, payout 0	Non-textbook / legitimate book
condition == 'mint' / 'poor'	42	Mint (1.25x), Poor (0.40x) multipliers	Good / other -> 1.0x multiplier
payout_in_credit == True	49	Bonus added to batch payout	Bonus not added
rejected_items > 4	54	5 rejected items -> AUDIT_REJECTED	<= 4 rejected -> checks payout
error_rate > 5 (scanner)	70	Error > 5 -> increments streak	Error <= 5 -> resets streak to 0



Figure 2: Branch Coverage Terminal Output (100% Branch Coverage)

3. Line Coverage vs. Branch Coverage Analysis

- **Difference:** Line coverage verifies if execution reached each line of code, while branch coverage verifies whether every control decision (if, elif, loop) evaluated to both True and False.
- **Why Line Coverage Alone Gives a False Sense of Security:** Line coverage ignores untested alternate paths in control blocks without an explicit else. For example, testing only payout_in_credit=True executes the credit bonus line, achieving 100% statement coverage for that block, but leaves the standard non-bonus logic completely untested. Similarly, testing only error rates > 5 increments the scanner streak, but fails to test whether clean scans (<= 5) properly reset the counter to 0.

4. Git Commit History

Commit Hash	Commit Message	Files Committed
b027b4c	Add code coverage test suite achieving 100% line coverage and terminal screenshot	CodeCoverageTest_bookstore_trade_in.py screenshots/code_coverage_analysis.png .gitignore
fa10ccb	Add branch coverage test suite achieving 100% branch coverage and terminal screenshot	BranchCoverageTest_bookstore_trade_in.py test_bookstore_trade.py screenshots/branch_coverage_analysis.png screenshots/baseline_initial_coverage.png
d7b2a17	Add comprehensive lab report deliverables (DOCX, PDF, and Markdown)	Lab_6_Report.md PES2UG24CS196_Hitha_Shree_Suresh.docx PES2UG24CS196_Hitha_Shree_Suresh.pdf